

Supplementary Appendix

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This appendix has been provided by the authors to give readers additional information about the work.

SUPPLEMENTAL APPENDIX

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Supplemental Methods

Patients needed to be eligible for treatment with the physician's choice of comparator (docetaxel, abiraterone acetate, or enzalutamide). Patients who were previously treated with a poly(adenosine diphosphate-ribose) polymerase (PARP) inhibitor, or chemotherapy in the castration-resistant setting, were excluded. Deleterious germline or somatic alterations in *BRCA1*, *BRCA2*, or *ATM* were determined by central testing of tissue and/or plasma using Foundation Medicine, Inc., next-generation sequencing assays,^{1,2} or through local testing; further details have been previously published.³ Whole blood samples were tested by Color Health to categorize alterations as somatic or germline.^{4,5}

Patients were randomized 2:1 to 600 mg oral rucaparib twice daily or physician's choice of: docetaxel (administered intravenously at a dose of 75 mg/m² every 3 weeks, up to a maximum of 10 cycles, in combination with prednisone or prednisolone 5 mg twice daily); abiraterone acetate (administered orally at a starting dose of 1000 mg once daily, in combination with prednisone 10 mg once daily); or enzalutamide (administered orally at a starting dose of 160 mg once daily). Randomization occurred via a central randomization procedure using an Interactive Response System.

For analysis of objective response rate, the independent radiographic review-evaluable population comprised patients with measurable disease at baseline per blinded, central independent radiographic review assessment; the investigator-evaluable population comprised patients with measurable disease at baseline per investigator assessment. The safety population included all patients who received ≥ 1 dose of protocol-specified treatment.

Imaging-based disease progression includes confirmed soft tissue disease progression and confirmed bone disease progression as per modified Response Evaluation Criteria in Solid Tumor Version 1.1/ Prostate Cancer Clinical Trials Working Group 3 (RECIST/PCWG3) criteria. Additional specific details are available in the protocol at NEJM.org.

According to the PCWG3⁶, imaging-based progression-free survival is the time interval from random assignment to the date when the first site of disease is found to progress (using a manifestation-specific definition of progression), or death, whichever occurs first.

To better understand the effect of therapy on an individual site of disease, the PCWG3 advises the date of progression in all specific sites be reported independently whether it is bone, nodes (pelvic or extra pelvic), visceral (e.g., lung, liver, adrenal, or CNS), or other. Reporting the proportion of patients who remain progression-free using radiographic assessment at fixed time points (e.g., 6 and 12 months) is also advised. The PCWG3 also emphasizes the importance of analytically and clinically validating the definitions of progression for each site of disease, as was performed for the bone scan progression end point proposed in the Prostate Cancer Working Group 2 that became part of the definition for imaging-based progression-free survival in the phase 3 registration trials of abiraterone (NCT00887198), enzalutamide (NCT01212991), and orteronel (NCT01193244). In all three of these trials, the local radiographic interpretations were highly concordant with the centralized interpretation.

Disease progression after initiation of most recent therapy was based on any of the following criteria: rise in prostate-specific antigen (PSA) (minimum of two consecutive rising levels, with an interval of ≥ 1 week between each determination); transaxial imaging (new or progressive soft tissue masses on CT or MRI scans as defined by modified RECIST; radionuclide bone scan (≥ 2 new metastatic lesions).

Tumor assessments were conducted at baseline and every 8 weeks for 24 weeks, then every 12 weeks, via computed tomography/magnetic resonance imaging and technetium-bone scans. Tumor assessments continued until radiographic disease progression was confirmed by independent radiographic review. PSA was evaluated at baseline and every 4 weeks thereafter for patients randomized to rucaparib or physician's choice of abiraterone acetate or

enzalutamide, and at baseline and every 3 weeks thereafter for patients randomized to docetaxel.

Patients had a 28-day follow-up visit after treatment discontinuation. A tumor assessment was performed for patients who had their last tumor assessment ≥ 8 weeks prior to treatment discontinuation (≥ 12 weeks if previous scan was after 24 weeks on study). Data on survival and subsequent treatment were collected every 12 weeks from the last dose of study drug until death, study closure, loss to follow-up, or consent withdrawal.

Safety was assessed by monitoring vital signs, laboratory testing, electrocardiogram, and physical examination. Adverse events of special interest (pneumonitis or similar events) were reported up to the 28-day follow-up visit. After the 28-day follow-up visit, myelodysplastic syndrome and acute myeloid leukemia, as well as treatment-related events meeting the criteria for a serious adverse event, were reported.

Imaging-based progression-free survival (PFS) was defined as the time from randomization to the date of first objective evidence of radiographic progression or death, whichever occurred first. Overall survival was defined as the date from randomization to the date of death.

Confirmed PSA response was defined as $\geq 50\%$ or $\geq 90\%$ reduction in PSA from baseline on ≥ 2 assessments conducted ≥ 3 weeks apart. Time to PSA progression was defined as the time from randomization to the date of a $\geq 25\%$ increase and absolute increase of ≥ 2 ng/mL above the nadir (or baseline value for patients who did not have a decline in PSA) in PSA, confirmed by a consecutive assessment conducted ≥ 3 weeks later. Duration of response was defined as the time from first response (modified RECIST in patients with visceral/nodal disease) until the first date of disease progression. Clinical benefit rate was defined as the proportion of patients without disease progression per modified RECIST/PCWG3 criteria.

PSA response in patients with both baseline and ≥ 1 postbaseline value were summarized with frequencies and proportion together with 95% confidence interval (CI) and compared between treatment groups using a Cochran-Mantel-Haenszel test. Imaging-based PFS, objective response rate, and duration of response were also assessed by the investigator. Stable disease was measured from the start of treatment until criteria for progression were met.

The primary endpoint was analyzed using the Cox proportional hazards model and tested using an ordered step-down multiple comparisons procedure in the BRCA subgroup and the intent-to-treat population. Imaging-based PFS by independent radiographic review in the BRCA subgroup was tested first at a one-sided 0.025 significance level. Analysis of imaging-based PFS by independent radiographic review in the intent-to-treat population was contingent upon a statistically significant result in the BRCA subgroup. For PFS, patients without documented progression were censored as of their last tumor assessment. Additional information about the censoring rules can be found in the Statistical Analysis Plan. Similar to the primary endpoint, analysis of key secondary endpoints (including overall survival and objective response rate by independent radiographic review) followed an ordered step-down procedure in the BRCA subgroup and intent-to-treat population. Once statistical significance was not achieved for one test, statistical significance could not be declared for all subsequent analyses in the ordered step-down procedure.

For treatment group comparisons, log-rank tests were stratified for imaging-based PFS by independent radiographic review and overall survival in the BRCA subgroup and intent-to-treat population, as well as for sensitivity analyses of imaging-based PFS by independent radiological review and imaging-based PFS by investigator. Stratification did not include the hepatic metastases variable due to the small number of patients. Log-rank tests were not stratified for all other time-to-event analyses. Statistical analyses were performed using SAS version 9.3 or higher (SAS Institute, Cary, NC, USA).

Dose-modification Criteria

Treatment interruptions or reductions were permitted in the event of unacceptable toxicity, and patients continued treatment until disease progression, loss to follow-up, unacceptable toxicity, or withdrawal of consent.

Rucaparib dose interruptions and/or reductions were permitted in the event of grade 3/4 toxicity (except for alopecia, nausea, vomiting, or diarrhea adequately controlled with systemic antiemetic/antidiarrheal medication), or grade 2 toxicity not adequately controlled by concomitant medications and/or supportive care. Dose reductions were permitted in 100 mg twice-daily decrements from 600 mg to 300 mg.

If a patient had persistent grade ≥ 3 anemia despite study drug interruption, and the anemia was considered related to the patient's underlying prostate cancer (e.g., patient had extensive bony disease or history of disease-related anemia), then resumption of study drug could be considered per investigator discretion.

If a patient experienced a grade 3 alanine aminotransferase (ALT) or aspartate aminotransferase (AST) elevation, rucaparib could be continued, in the absence of other signs of liver dysfunction, provided that levels did not continue to rise, and bilirubin was less than the upper limit of normal. If levels did not decline within 2 weeks or continued to rise, treatment interruption and resolution to grade ≤ 2 was required before resuming rucaparib at the current or a reduced dose. If a patient experienced a grade 4 ALT/AST elevation, rucaparib was interrupted until resolution to grade ≤ 2 , and resumed at a reduced dose.

If new or worsening unexplained pulmonary symptoms suggestive of pneumonitis (including, but not limited to, dyspnea) occurred, a deterioration of pulmonary function was observed, and/or radiologic abnormality was detected in the lungs, and this event occurred in the absence of any clear diagnosis, a diagnostic workup (including high resolution CT scan) in consultation with a

pulmonologist was performed to rule out pneumonitis. During this time, treatment with rucaparib may be interrupted or continued per investigator discretion. Following investigation, if pneumonitis was not confirmed, treatment with rucaparib would resume/continue as deemed appropriate by the investigator and in accordance with the study protocol directions for management of adverse events. Any confirmed cases of pneumonitis were treated per medical judgement and institutional guidelines; if the event resolved, retreatment with rucaparib could be considered at the current or a reduced dose.

Dose modifications for all comparator therapies were in accordance with the approved Prescribing Information or according to institutional guidelines or local dose-reduction protocols.

Protocol Violations

Major protocol violations included non-protocol study drug dosing or non-compliance in 14 patients, efficacy assessments not completed in 12 patients, inclusion criteria not met in 4 patients, screening/baseline assessments missing for 3 patients, and concomitant anti-cancer treatment use in 1 patient.

References for Supplemental Methods

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4. Crawford B, Adams SB, Sittler T, et al. Multi-gene panel testing for hereditary cancer predisposition in unsolved high-risk breast and ovarian cancer patients. *Breast Cancer Res Treat* 2017;163:383-90.
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6. Scher HI, Morris MJ, Stadler WM, et al. Prostate Cancer Clinical Trials Working Group 3. Trial design and objectives for castration-resistant prostate cancer: Updated recommendations from the Prostate Cancer Clinical Trials Working Group 3. *J Clin Oncol* 2016;34:1402-18.

Table S1. Representativeness of Study Participants

Category	Commentary
Disease, problem, or condition under investigation	Metastatic castration-resistant prostate cancer (mCRPC) in men with deleterious <i>BRCA1</i> , <i>BRCA2</i> , or <i>ATM</i> alterations who had not received chemotherapy in this setting.
Special considerations related to	
Sex and gender	Prostate cancer occurs in males or females assigned male at birth. Enrollment by sex and gender in TRITON3 was consistent with what would be expected in this patient population.
Age	The risk of developing prostate cancer and mCRPC increases with age.¹ According to the National Cancer Institute, approximately 60% of prostate cancer cases in the United States are diagnosed in men who are ≥65 years old.² Worldwide, as many as three-quarters of cases occur in men who are ≥65 years old.³ Additionally, older men are more likely to experience mCRPC.⁴ In TRITON3, the median age of participants in the rucaparib and physician's choice groups was 70 and 71 years, respectively. In addition, 69% and 76% of patients were ≥65 years old in the rucaparib and physician's choice groups, respectively.

	In TRITON3, the median age and proportion of the population with prostate cancer that is ≥ 65 years old is consistent with what would be expected in the general population.
Race or ethnic group	Men of African descent are disproportionately affected by prostate cancer.⁵ In the United States, men of African descent are 60% more likely to develop prostate cancer compared with European-American men and are more than twice as likely to die from the disease.⁶ Prostate cancer mortality is highest among men of African origin in the Caribbean (Afro-Caribbean men).⁵ The increased death rate in men of African descent may be attributable to biological differences, poor access to health care, socioeconomic, and other factors.⁷ In TRITON3, where reported, only 4% and 3% of patients in the rucaparib and physician's choice groups, respectively, were men of African descent. Given that prostate cancer disproportionately affects men of African descent, this patient population was underrepresented in the TRITON3 study. Some countries do not permit collection of race or ethnic data; therefore, 20% of these data are reported "missing" in both the rucaparib and physician's choice group in TRITON3.

Geography	In 2020, prostate cancer was the second most frequent cancer and the fifth leading cause of cancer death worldwide in men.⁸ The highest incidence of prostate cancer is found in Northern and Western Europe, the Caribbean, Australia/New Zealand, Northern America, and Southern Africa. The lowest incidence of prostate cancer is found in Asia and Northern Africa.⁸ Regional patterns of mortality attributable to prostate cancer do not necessarily follow the incidence of prostate cancer. For example, the Caribbean, sub-Saharan Africa, and Micronesia/Polynesia have the highest mortality rates in the world (age-standardized rate per 100,000 men) despite having a lower prevalence than Northern and Western Europe.⁸ In countries with a high/very high human development index (HDI), prostate cancer is typically the second most common cancer behind lung carcinoma in men. In low/medium HDI countries, prostate cancer is typically the most common cancer (age-standardized rate per 100,000 males).⁸ Patients from 12 countries (including Australia, Canada, Israel, the United States, and Western Europe) were enrolled into the TRITON3 study.
Other considerations	There are a limited number of publications describing the incidence of pathogenic or likely pathogenic germline and

	somatic alterations in <i>BRCA1</i>, <i>BRCA2</i>, or <i>ATM</i> in localized and advanced prostate cancer. A meta-analysis⁹ found that the rate of germline <i>BRCA1</i> and <i>BRCA2</i> pathogenic/non-pathogenic alterations was 0.5% and 3.5%, respectively. The rate of somatic <i>BRCA1</i> and <i>BRCA2</i> pathogenic and non-pathogenic alterations was 5.7% and 3.2%, respectively. The frequency of pathogenic germline <i>ATM</i> alterations in advanced prostate cancer is estimated to be 1.6–2.0%.¹⁰⁻¹² In addition, germline or somatic alterations in <i>ATM</i> are present in 5–8% of castration-resistant prostate cancer, an incidence that may be nearly twice as frequent vs. localized prostate cancer.^{10,13,14} In TRITON3, the inclusion of men with pathogenic or suspected pathogenic germline or somatic <i>BRCA1</i>, <i>BRCA2</i>, or <i>ATM</i> alterations is consistent with prior reports. Additional population-based studies are needed to get a better sense of the incidence of pathogenic or likely pathogenic germline and somatic alterations in <i>BRCA1</i>, <i>BRCA2</i>, and <i>ATM</i> in localized and advanced prostate cancer.
Overall representativeness of this trial	The gender, age, and genomic features of participants in TRITON3 were generally representative of the population at risk for prostate cancer; however, men of African descent

	were underrepresented in TRITON3 relative to the general population with advanced prostate cancer.
References 1. Leitzmann MF, Rohrmann S. Risk factors for the onset of prostatic cancer: age, location, and behavioral correlates. <i>Clin Epidemiol</i> 2012;4:1-11. 2. National Cancer Institute: Cancer Stat Facts: Prostate Cancer. 2022, Bethesda: National Cancer Institute. 3. Quinn M, Babb P. Patterns and trends in prostate cancer incidence, survival, prevalence and mortality. Part I: international comparisons. <i>BJU international</i> 2002;90:162-73. 4. Ged Y, Horgan AM. Management of castrate-resistant prostate cancer in older men. <i>J Geriatr Oncol</i> 2016;7:57-63. 5. McGinley KF, Tay KJ, Moul JW. Prostate cancer in men of African origin. <i>Nat Rev Urol</i> 2016;13:99-107. 6. Powell IJ. Epidemiology and pathophysiology of prostate cancer in African-American men. <i>J Urol</i> 2007;177:444-9. 7. Hanson HA, Martin C, O'Neil B, et al. The relative importance of race compared to health care and social factors in predicting prostate cancer mortality: a random forest approach. <i>J Urol</i> 2019;202:1209-16. 8. Sung H, Ferlay J, Siegel RL, et al. Global Cancer Statistics 2020: GLOBOCAN Estimates of Incidence and Mortality Worldwide for 36 Cancers in 185 Countries. <i>CA Cancer J Clin</i> 2021;71:209-49. 9. Shao C, Wan J, Lam FC, et al. A comprehensive literature review and meta-analysis of the prevalence of pan-cancer BRCA mutations, homologous recombination repair gene mutations, and homologous recombination deficiencies. <i>Environ Mol Mutagen</i> 2022;63:308-16. 10. Pritchard CC, Mateo J, Walsh MF, et al. Inherited DNA-repair gene mutations in men with metastatic prostate cancer. <i>N Engl J Med</i> 2016;375:443-53. 11. Nicolosi P, Ledet E, Yang S, et al. Prevalence of germline variants in prostate cancer and implications for current genetic testing guidelines. <i>JAMA Oncol</i> 2019;5:523-28. 12. Castro E, Romero-Laorden N, Del Pozo A, et al. PROREPAIR-B: A prospective cohort study of the impact of germline DNA repair mutations on the outcomes of patients with metastatic castration-resistant prostate cancer. <i>J Clin Oncol</i> 2019;37:490-503. 13. Robinson D, Van Allen EM, Wu YM, et al. Integrative clinical genomics of advanced prostate cancer. <i>Cell</i> 2015;161:1215-28. 14. Abida W, Cyrta J, Heller G, et al. Genomic correlates of clinical outcome in advanced prostate cancer. <i>Proc Natl Acad Sci U S A</i> 2019;116:11428-36.	

Table S2. Sensitivity Analyses for Imaging-based Progression-Free Survival by Independent Radiographic Review.

	BRCA Subgroup		Intent-to-Treat Population	
	Rucaparib (N = 201)	Physician's Choice (N = 101)	Rucaparib (N = 270)	Physician's Choice (N = 135)
Median imaging-based PFS by IRR including clinical progression, months (95% CI)	9.1 (8.3–11.1)	5.8 (4.4–7.8)	8.3 (8.1–9.1)	5.8 (4.5–7.6)
Log-rank P value	0.0001		0.0022	
Hazard ratio (95% CI)	0.57 (0.42–0.76)		0.68 (0.54–0.87)	
Median imaging-based PFS by IRR including PSA progression, months (95% CI)	8.2 (7.4–9.2)	5.6 (3.8–7.1)	8.1 (7.0–8.2)	5.4 (3.9–6.1)
Log-rank P value	0.0001		0.0017	
Hazard ratio (95% CI)	0.58 (0.43–0.77)		0.68 (0.54–0.87)	
Median imaging-based PFS by IRR including scans after subsequent anti-cancer therapy, months (95% CI)	11.2 (8.9–12.0)	7.1 (5.6–8.4)	9.4 (8.3–11.1)	7.1 (5.7–8.4)
Log-rank P value	0.0003		0.0025	
Hazard ratio (95% CI)	0.58 (0.43–0.78)		0.68 (0.53–0.87)	

The widths of the 95% confidence intervals were not adjusted for multiplicity and cannot be used in place of hypothesis testing. BRCA denotes *BRCA1* and *BRCA2*, CI confidence interval,

IRR independent radiographic review, PFS progression-free survival, and PSA prostate-specific antigen.

Table S3. Intent-to-Treat Population Censored for Imaging-based Progression-Free Survival in Patients who Received at least One Subsequent Anticancer Therapy by Independent Radiographic Review.

	Rucaparib (N = 56)	Physician's Choice		
		Docetaxel (N = 11)	Second- generation ARPI (N = 7)	Total (N = 18)
End of therapy reason, n (%)				
Adverse Event	9 (16)	4 (36)		4 (22)
Other	-	3 (27)		3 (17)
Patient Withdrawal of Consent for Treatment	2 (4)	-	-	-
Patient Withdrawal of Consent from study	-	-	1 (14)	1 (6)
Physician decision	5 (9)	2 (18)	-	2 (11)
Progressive disease	39 (70)	2 (18)	6 (86)	8 (44)
Protocol deviation	1 (2)	-	-	-

Table S4. Confirmed Objective Response in Patients with Measurable Disease by Independent Radiographic Review at Baseline.*

	BRCA Subgroup		Intent-to-Treat Population		ATM Subgroup	
	Rucaparib (N = 82)	Physician's Choice (N = 41)	Rucaparib (N = 106)	Physician's Choice (N = 55)	Rucaparib (N = 24)	Physician's Choice (N = 14)
Confirmed objective response rate by IRR*, %	45	17	35	16	0	14
95% CI	34.1–56.5	7.2–32.1	25.9–44.8	7.8–28.8	0–14.2	1.8–42.8
Best overall confirmed response, no. (%)						
Complete response	10 (12)	2 (5)	10 (9)	2 (4)	0	0
Partial response	27 (33)	5 (12)	27 (25)	7 (13)	0	2 (14)
Stable disease	34 (41)	21 (51)	50 (47)	28 (51)	16 (67)	7 (50)
Progressive disease	10 (12)	10 (24)	15 (14)	14 (25)	5 (21)	4 (29)
Not evaluable	1 (1)	3 (7)	4 (4)	4 (7)	3 (13)	1 (7)

*Per modified RECIST/PCWG3 criteria. The widths of the 95% confidence intervals were not adjusted for multiplicity and cannot be used in place of hypothesis testing. BRCA denotes *BRCA1* and *BRCA2*, CI confidence interval, IRR independent radiographic review, and RECIST/PCWG3 Response Evaluation Criteria in Solid Tumor Version 1.1/Prostate Cancer Clinical Trials Working Group 3.

Table S5. Duration of Response in Patients with Confirmed Response by Independent Radiographic Review.*

	BRCA Subgroup		Intent-to-Treat Population		ATM Subgroup	
	Rucaparib (n=48)	Physician's Choice (n=9)	Rucaparib (n=49)	Physician's Choice (n=11)	Rucaparib (n=1)	Physician's Choice (n=2)
Median duration of response, months (95% CI)	7.5 (6.5–12.0)	7.4 (3.5–NE)	7.4 (6.5–12.0)	7.4 (4.4–14.5)	4.8 (NE–NE)	11.0 (7.4–14.5)
Hazard ratio (95% CI)	0.75 (0.27–2.10)		0.86 (0.36–2.04)		5.90 (0.07–535.35)	

*Includes patients with nonmeasurable disease at baseline who had a complete response per IRR. The widths of the 95% confidence intervals were not adjusted for multiplicity and cannot be used in place of hypothesis testing. BRCA denotes *BRCA1* and *BRCA2*, CI confidence interval, and IRR independent radiographic review.

Table S6. Clinical Benefit Rate by Independent Radiographic Review.

	BRCA Subgroup		Intent-to-Treat Population		ATM Subgroup	
	Rucaparib (N=201)	Physician's Choice (N=101)	Rucaparib (N=270)	Physician's Choice (N=135)	Rucaparib (N=69)	Physician's Choice (N=34)
Six-month clinical benefit rate by IRR, % (n/N)	63 (126/200)	23 (22/97)	58 (155/269)	25 (33/130)	42 (29/69)	33 (11/33)
95% CI	55.9–69.7	14.8–32.3	51.5–63.6	18.2–33.8	30.2–54.5	18.0–51.8
Twelve-month clinical benefit rate by IRR, % (n/N)	31 (54/177)	4 (3/85)	24 (60/246)	5 (6/118)	9 (6/69)	9 (3/33)
95% CI	23.8–37.9	0.7–10.0	19.2–30.3	1.9–10.7	3.3–18.0	1.9–24.3

The widths of the 95% confidence intervals were not adjusted for multiplicity and cannot be used in place of hypothesis testing.

BRCA denotes *BRCA1* and *BRCA2*, CI confidence interval, and IRR independent radiographic review.

Table S7. PSA Response Rates in the BRCA Subgroup and Intent-to-Treat Population.

	BRCA Subgroup		Intent-to-Treat Population	
	Rucaparib (N = 201)	Physician's Choice (N = 101)	Rucaparib (N = 270)	Physician's Choice (N = 135)
PSA50 response rate, %	55	27	42	27
95% CI	47.6–61.7	18.4–36.5	35.9–48.0	19.4–35.0
PSA90 response rate, %	22	4	17	4
95% CI	16.8–28.8	1.1–9.8	12.4–21.7	1.2–8.4
Median time to PSA progression, months (95% CI)	6.6 (5.9–7.7)	3.8 (3.1–4.5)	5.7 (4.6–6.5)	3.6 (3.2–4.5)
Hazard ratio (95% CI)	0.52 (0.38–0.70)		0.63 (0.49–0.81)	

The widths of the 95% confidence intervals were not adjusted for multiplicity and cannot be used in place of hypothesis testing. BRCA denotes *BRCA1* and *BRCA2*, CI confidence interval, PSA prostate-specific antigen, PSA50 confirmed PSA response rate of $\geq 50\%$ decrease, and PSA90 confirmed PSA response rates of $\geq 90\%$ decrease.

Table S8. Investigator-Assessed Efficacy Endpoints in the BRCA Subgroup and Intent-to-Treat Population.

	BRCA Subgroup		Intent-to-Treat Population	
All Patients	Rucaparib (N = 201)	Physician's Choice (N = 101)	Rucaparib (N = 270)	Physician's Choice (N = 135)
Median imaging-based PFS by investigator, months (95% CI)	11.2 (10.7–14.2)	8.2 (6.3–8.8)	10.9 (9.5–13.6)	8.4 (7.1–10.4)
Hazard ratio (95% CI)	0.55 (0.40–0.76)		0.66 (0.50–0.87)	
Patients with measurable disease per investigator at baseline	Rucaparib (n=93)	Physician's choice (n=50)	Rucaparib (n=116)	Physician's choice (n=68)
Confirmed objective response rate by investigator, %	45	28	36	24
95% CI	34.8–55.8	16.2–42.5	27.5–45.6	14.1–35.4
Patients with confirmed response per investigator*	Rucaparib (n=47)	Physician's choice (n=15)	Rucaparib (n=47)	Physician's choice (n=17)
Median duration of response, months (95% CI)	9.5 (6.5–15.0)	5.8 (5.3–11.0)	9.5 (6.5–15.0)	7.4 (5.3–11.0)
Hazard ratio (95% CI)	0.58 (0.27–1.26)		0.57 (0.27–1.22)	

*Includes patients with nonmeasurable disease at baseline who had a complete response per investigator. The widths of the 95% confidence intervals were not adjusted for multiplicity and cannot be used in place of hypothesis testing. BRCA denotes *BRCA1* and *BRCA2*, CI confidence interval, DOR duration of response, IRR independent radiographic review, ORR objective response rate, PFS progression-free survival.

Figure S1. Ordered Step-Down Procedure. Following a protocol amendment on February 18, 2022, overall survival was elevated to the first secondary endpoint in the step-down analysis procedure. BRCA denotes *BRCA1* and *BRCA2*, and ITT intent-to-treat.

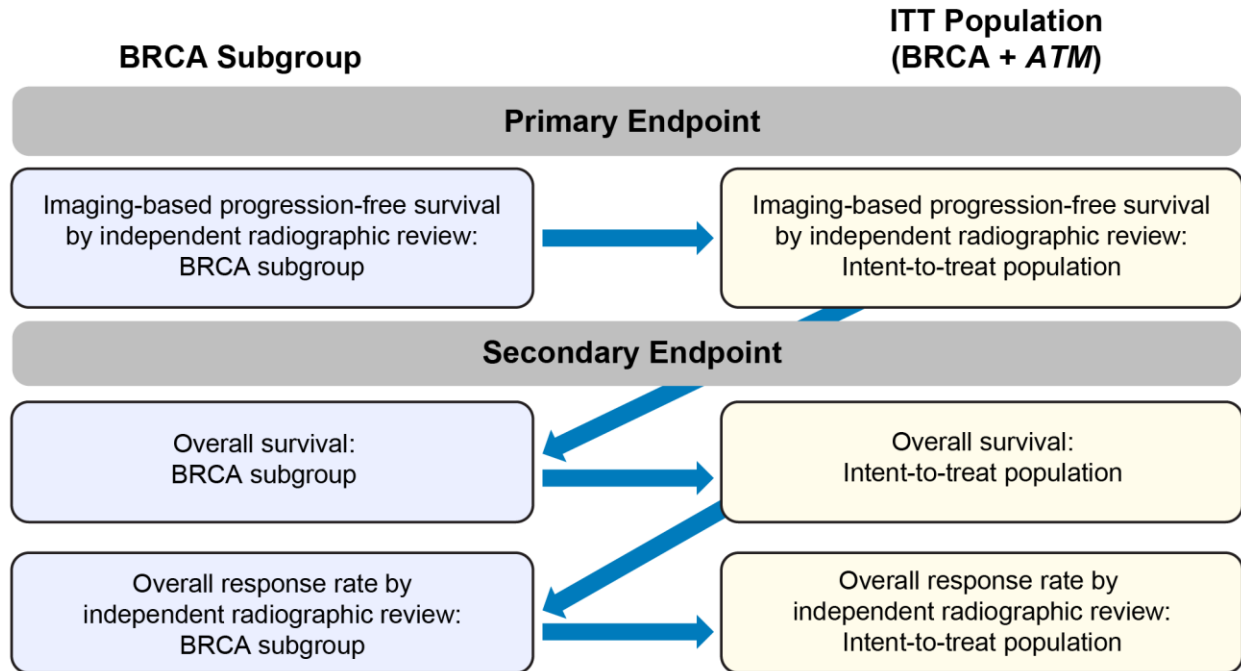


Figure S2. CONSORT Diagram. *Other reasons for screen fail included no confirmed adenocarcinoma of prostate, life expectancy <6 months, active malignancy, prior poly(adenosine diphosphate-ribose) polymerase inhibitor treatment, ineligible for treatment with physician's choice treatments, prior chemotherapy, no disease progression after prior treatment, or failure to meet other eligibility criteria. BRCA denotes *BRCA1* and *BRCA2*, ECOG PS Eastern Cooperative Oncology Group performance status, ICF informed consent form, and mCRPC metastatic castration-resistant prostate cancer.

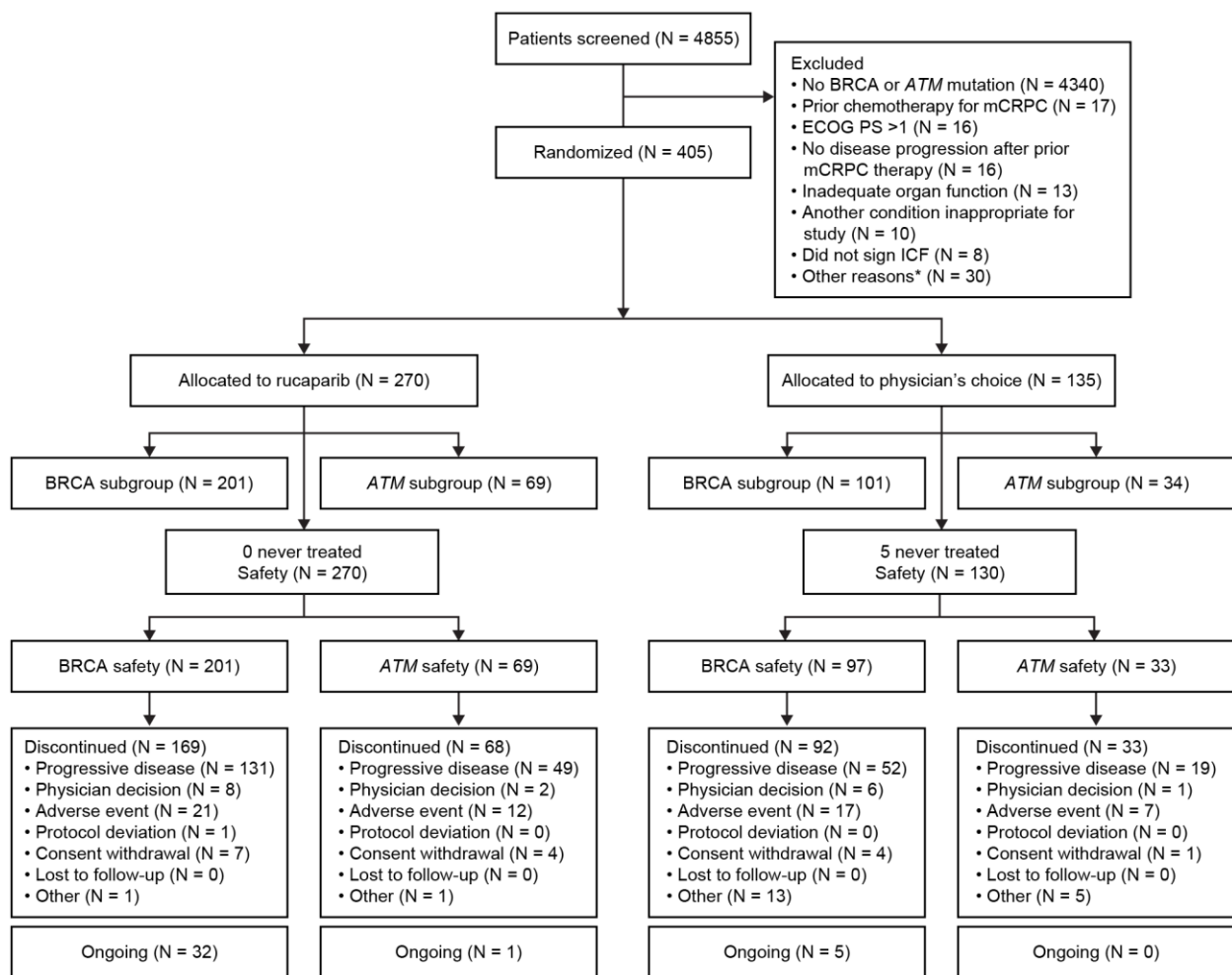
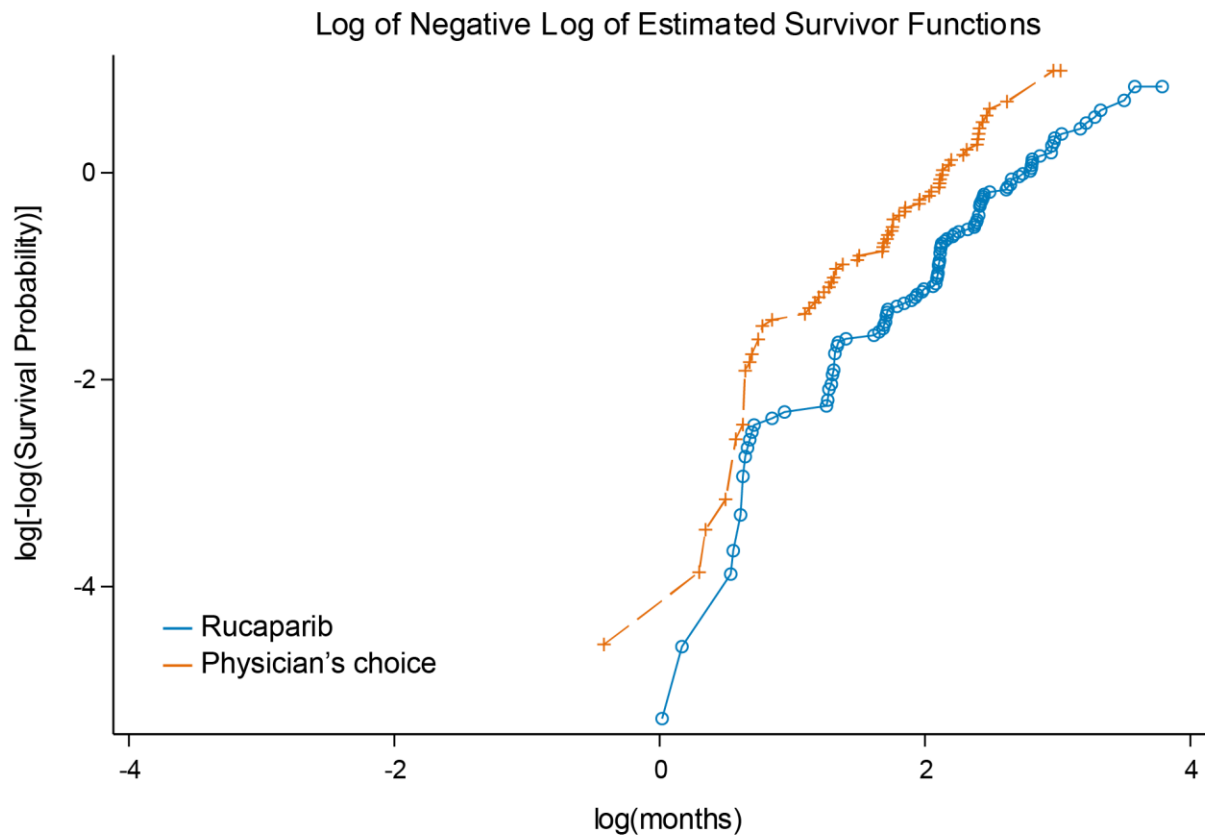


Figure S3. Plots of the Log of the Cumulative Hazard.

Log-log plots of the time to imaging-based progression-free survival by independent radiographic review in the rucaparib and physician's choice groups in the BRCA subgroup (A) and intent-to-treat population (B).

A. BRCA subgroup



B. Intent-to-treat population

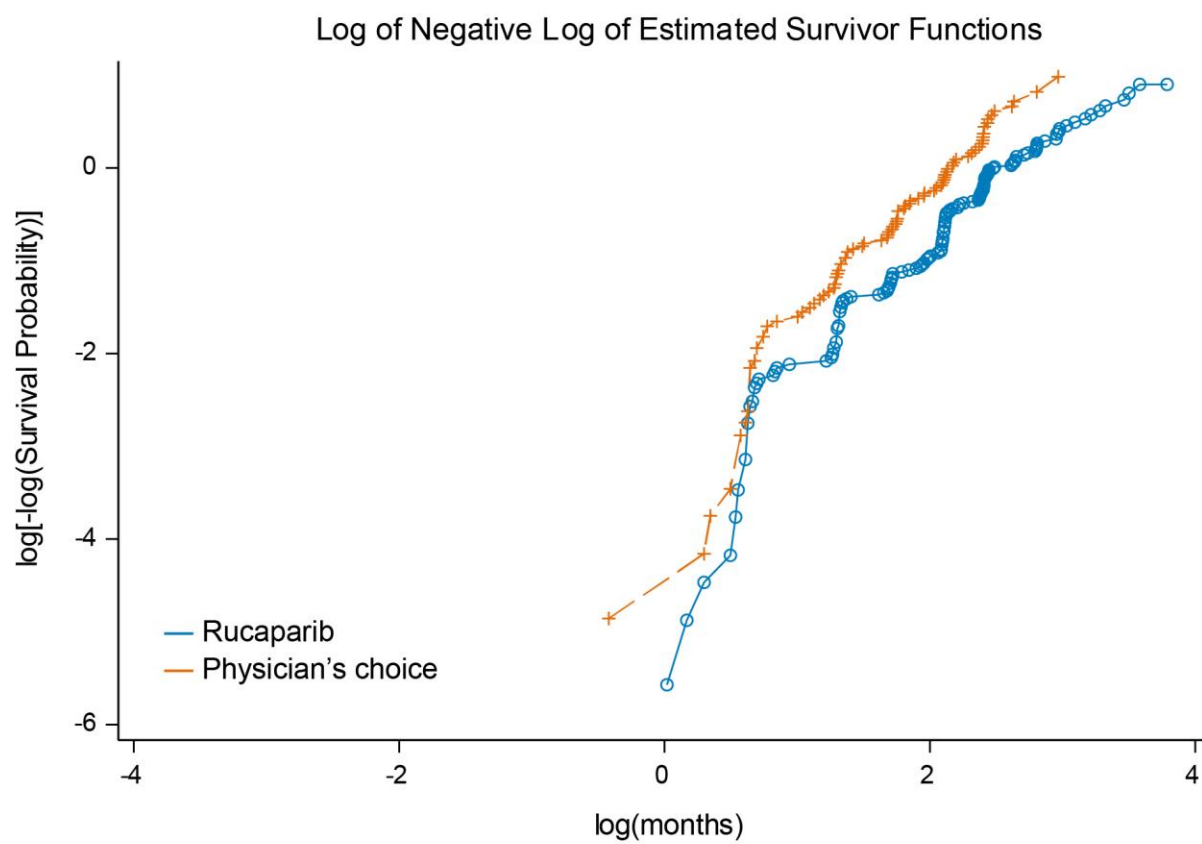
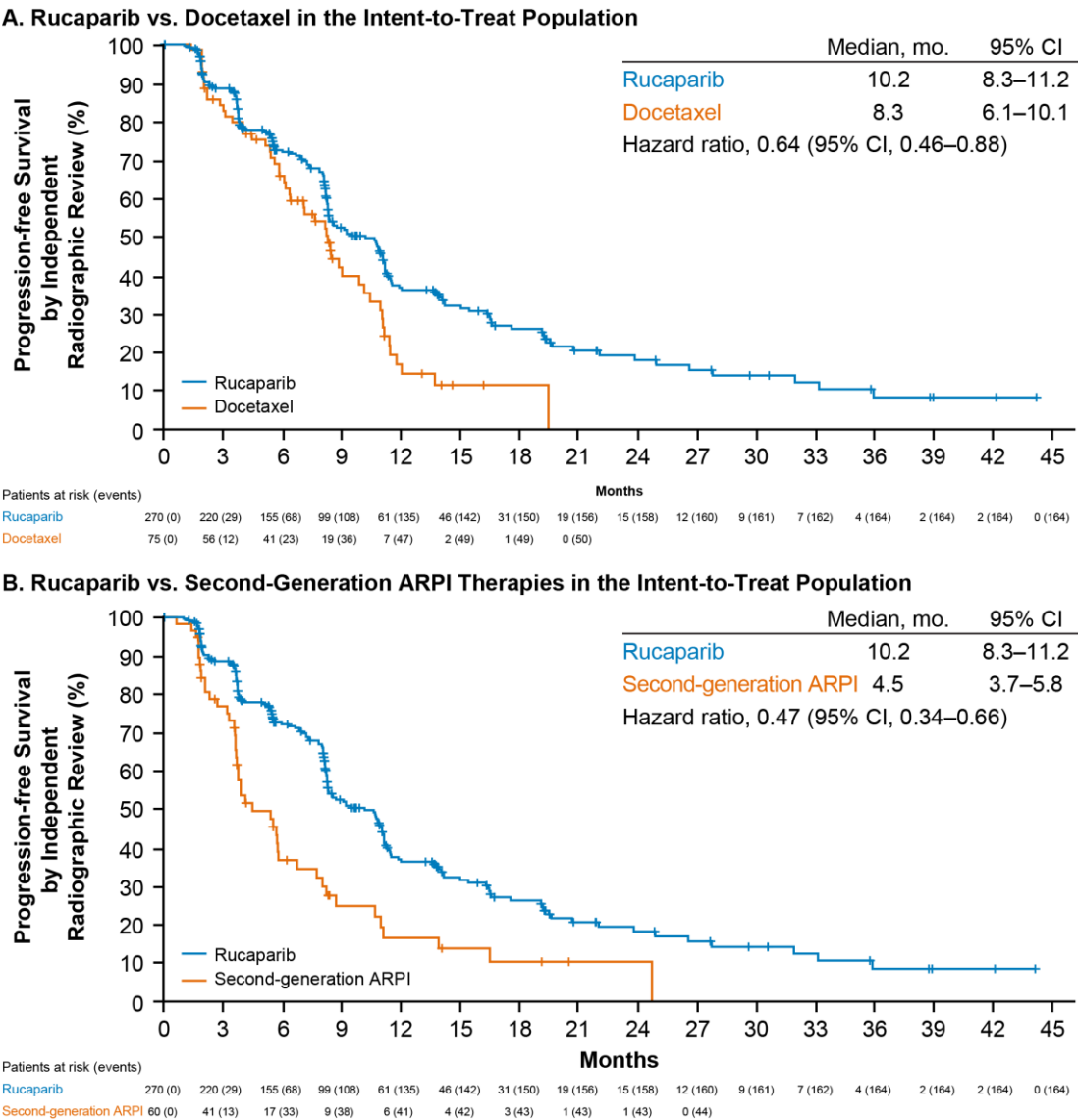
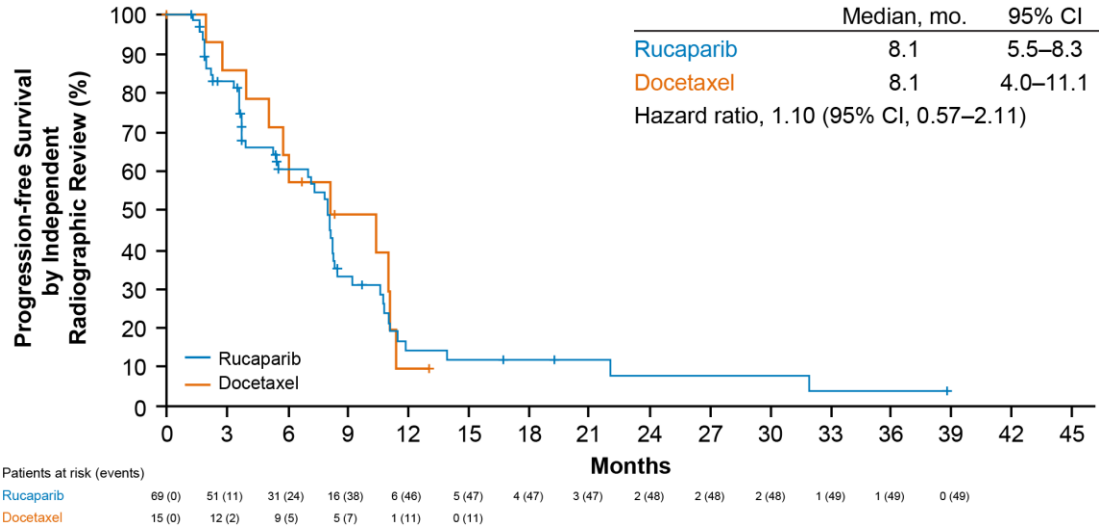


Figure S4. Kaplan-Meier Curves for Imaging-based Progression-free Survival by Independent Radiographic Review Based on Physician's Choice of Therapy.

The widths of the 95% confidence intervals were not adjusted for multiplicity and cannot be used in place of hypothesis testing. ARPI denotes androgen receptor pathway inhibitor, BRCA *BRCA1* and *BRCA2*, and CI confidence interval.



C. Rucaparib vs. Docetaxel in the ATM Subgroup



D. Rucaparib vs. Second-Generation ARPI Therapies in the ATM Subgroup

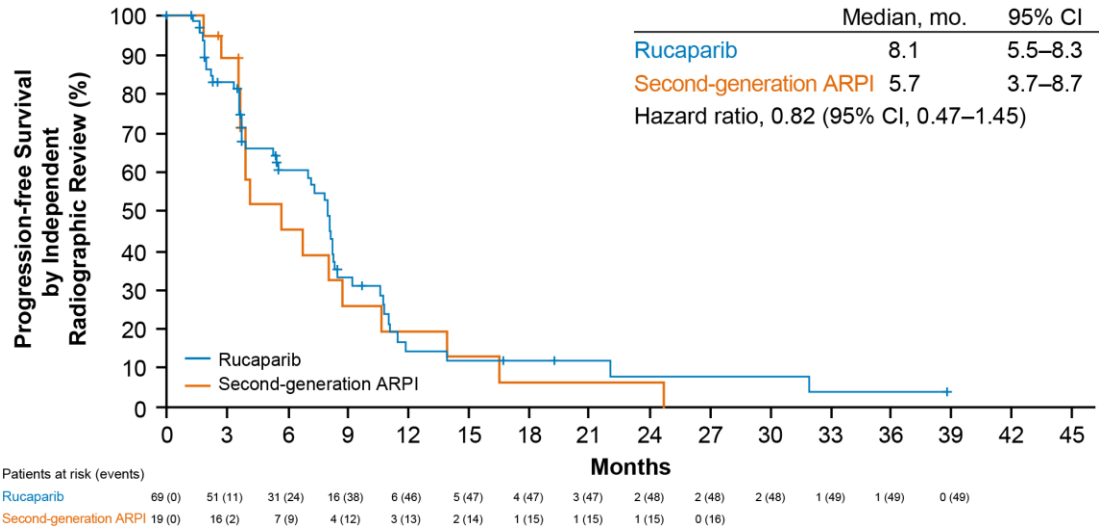


Figure S5. Imaging-based Progression-free Survival for Prespecified Subgroups Within the Intent-to-Treat Population by Independent Radiographic Review.

The widths of the 95% confidence intervals were not adjusted for multiplicity and cannot be used in place of hypothesis testing. ARPI denotes androgen pathway inhibitor, BRCA *BRCA1* and *BRCA2*, CI confidence interval, ITT intent-to-treat, and US United States.

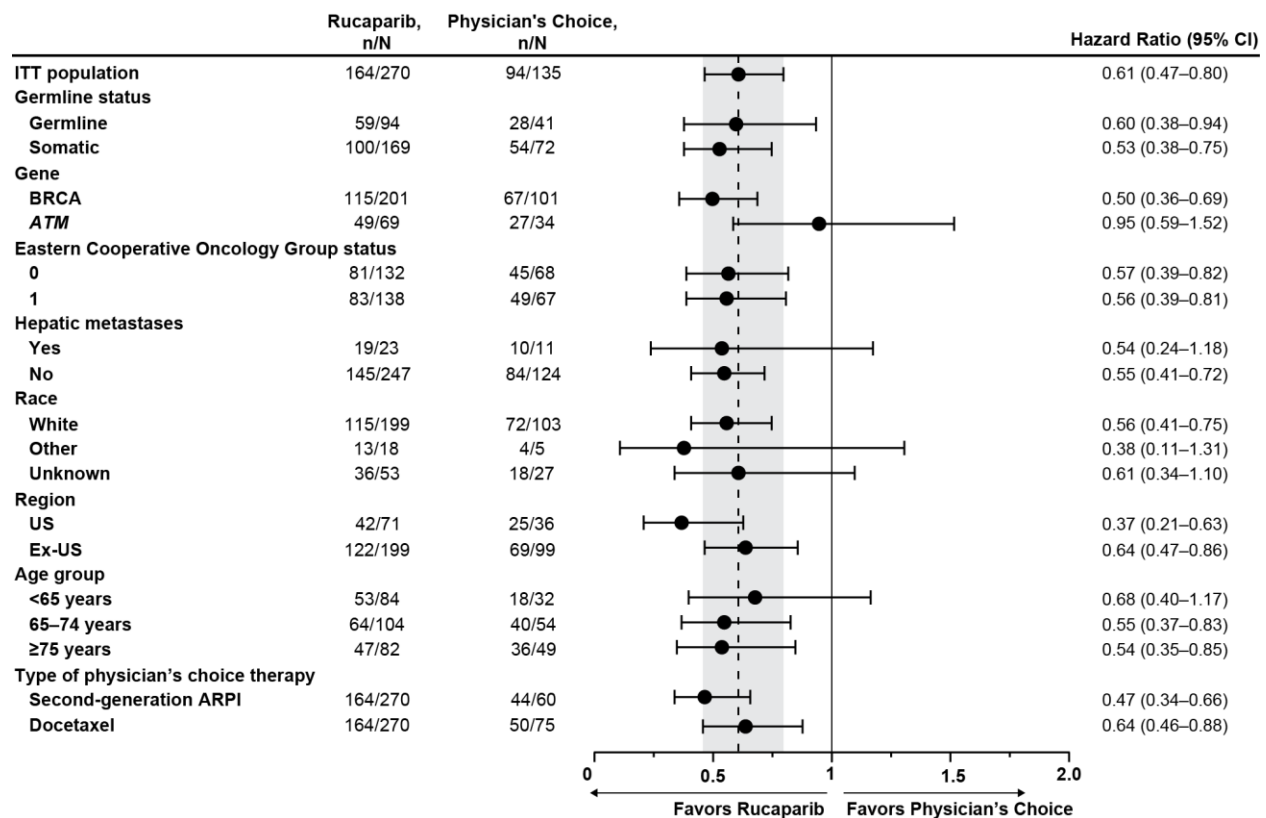


Figure S6. Kaplan-Meier Curves for Interim Overall Survival.

The widths of the 95% confidence intervals were not adjusted for multiplicity and cannot be used in place of hypothesis testing. BRCA denotes *BRCA1* and *BRCA2* and CI confidence interval.

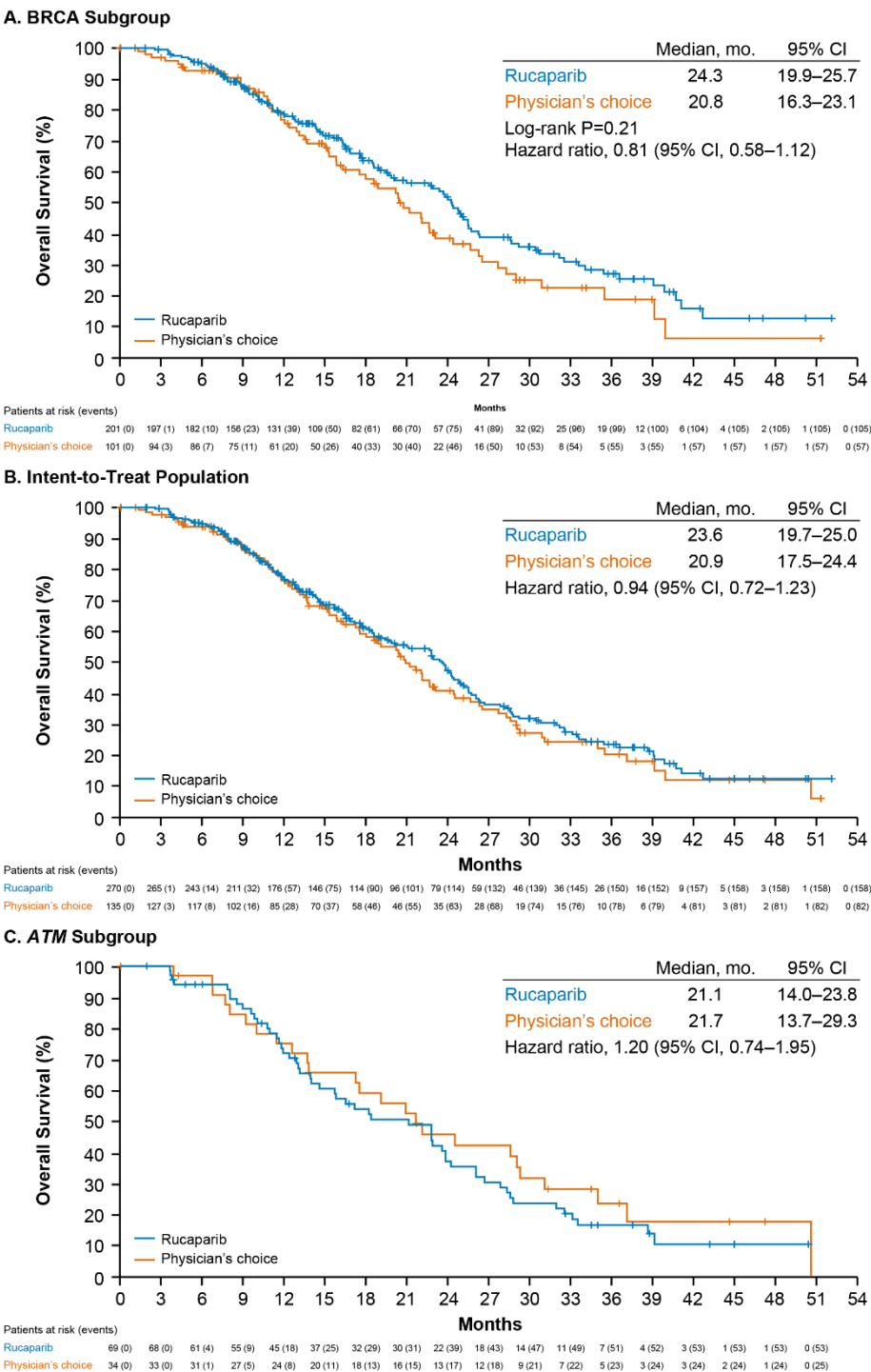
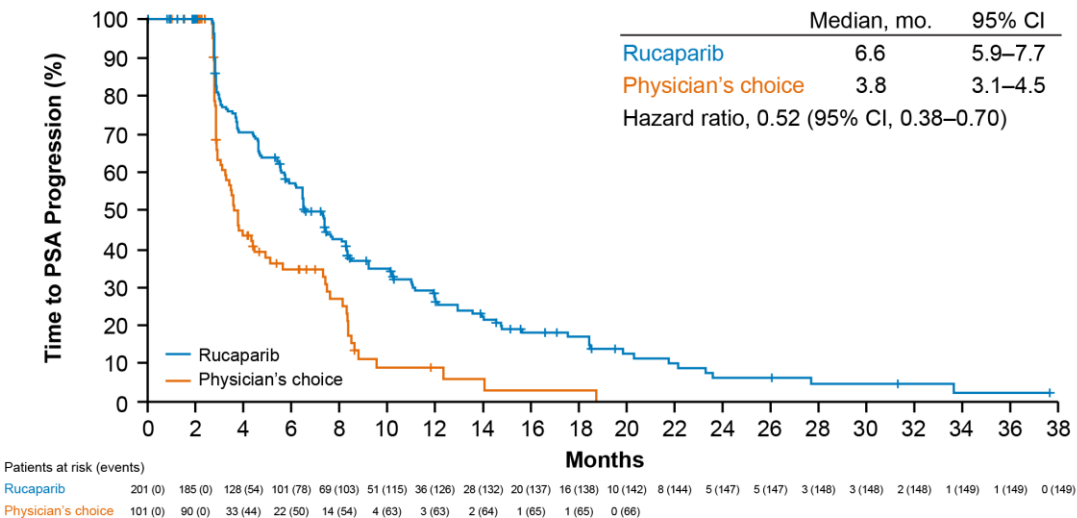


Figure S7. Kaplan-Meier Curves for Time to PSA Progression.

The widths of the 95% confidence intervals were not adjusted for multiplicity and cannot be used in place of hypothesis testing. BRCA denotes *BRCA1* and *BRCA2*, CI confidence interval, and PSA prostate-specific antigen.

A. BRCA Subgroup



B. Intent-to-Treat Population

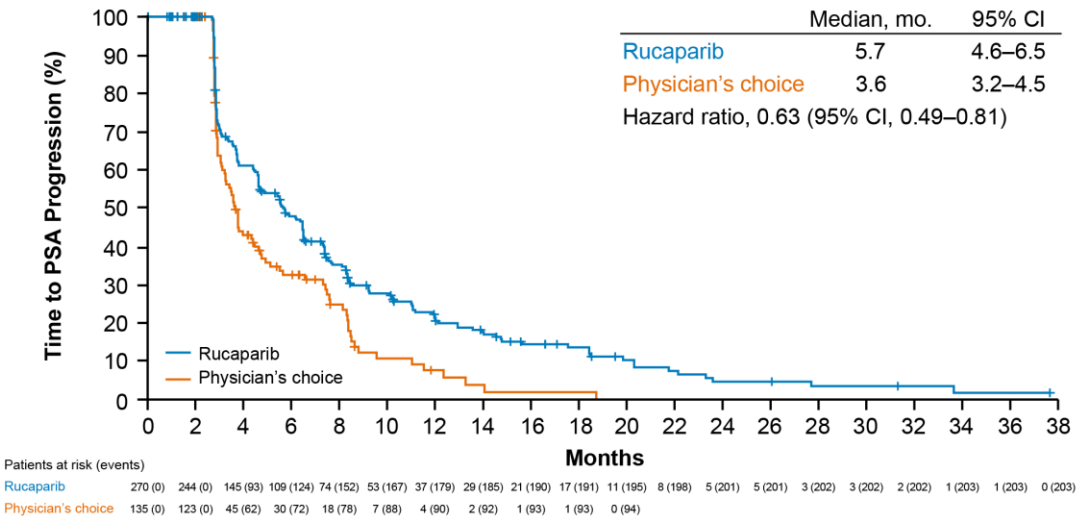
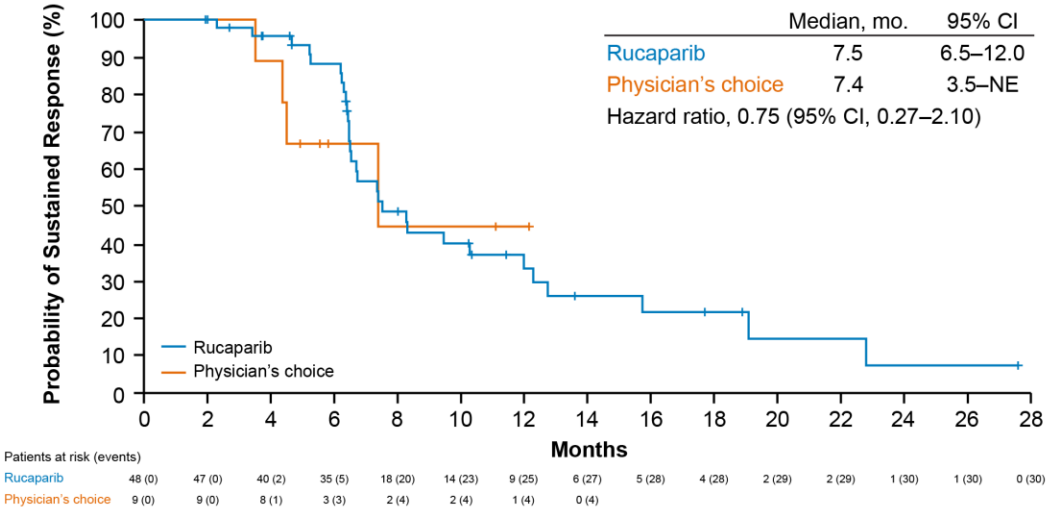


Figure S8. Kaplan-Meier Curves for Duration of Response by Independent Radiographic Review. The widths of the 95% confidence intervals were not adjusted for multiplicity and cannot be used in place of hypothesis testing. BRCA denotes *BRCA1* and *BRCA2*, CI confidence interval, and NE not evaluable.

A. BRCA Subgroup



B. Intent-to-Treat Population

